[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SANDER, PAD, PNEUMATIC, PORTABLE (ORBITAL MOTION)

This Commercial Item Description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. Scope. This commercial item description covers portable, pneumatic, orbital motion, pad sanders, used for finish sanding, feather edging, and blending.

2. Salient Characteristics.

2.1 Design. The sander shall be of the orbital motion type, be similar to Figure 1, and conform to the requirements specified in Table I. The sanders shall be equipped with two effective, hand-operated holders for securing the abrasive sheets to the pad. The sander shall operate properly at the manufacturers recommended air operating pressure range.

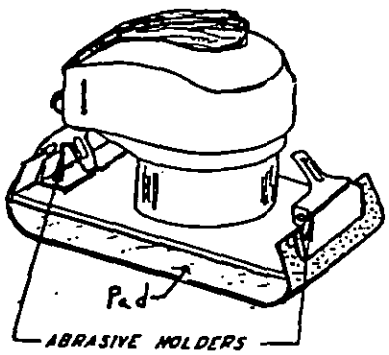


Figure 1. Sander.

Table I

National Stock Number	No Load Speed Orbits per minute (min.)	Air Consumption at No Load CFM (max.)	Abrasive Area Square inch	Paper Size (nominal)
5130-00-606-9694	6000 - 9000	18	15 - 20	3" x 8"

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5130

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2.2 Orbital motion: The sander shall approximately subscribe an orbit with a minimum diameter of 5/32 inch.

2.3 Back-up pad: Each sander shall contain a rectangular shaped pad made of neoprene rubber at lease 1/4 inch thick.

2.4 Materials: The materials used in the construction of all components shall be capable of completing the examination and tests required herein. All components shall be new and unused. Repaired or reconditioned components shall not be acceptable. Cadmium is not an acceptable plating.

2.5 Motor: The sander shall be equipped with a rotary vane-type pneumatic motor.

2.6 Throttle: The sander shall fit conveniently into the palm of a hand and be equipped with a palm-driven lever type throttle that shall automatically close the air inlet valve when hand pressure is removed.

2.7 Lubrication: Parts requiring lubrication shall be protected from the entrance of foreign matter and loss of lubricant. The sander shall provide for the delivery of lubricant in a manner that will assure proper operation.

2.8 Air inlet connection: The sander shall have a female, 1/4 inch size, NPT or NPTF, air inlet connection. The inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall prevent solid particles in the air supply from entering the tool. The filter shall be easy to remove, clean, and replace.

2.9 Bearings: The rotor shaft shall be provided with ball bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. If open type bearings are used, they shall be suitably housed and adequately sealed to prevent leakage of lubricant and entrance of dirt and dust. Sleeve bearings are not acceptable.

2.10 Manual: One copy of a technical manual consisting of the manufacturer's standard commercial instruction and parts list shall be furnished with each tool.

2.11 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

3. Regulatory Requirements: In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

3.1 Metric products: Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

4. Quality assurance provisions.

4.1 Responsibility for inspection: Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the

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contractor may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services adhere to prescribed requirements.

4.2 Inspection: It shall be determined that the sample units meet all salient characteristics stated in the requirements section of this document. Any sample containing one or more defects shall be rejected, and if the number of rejected units exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Test conditions: All tests shall be conducted with an air supply of 90 psig (620 kPa) or at the manufacturers recommended pressure measured as close to the tool as practicable. The test tool shall be connected to the air supply line by means of a 10 foot length of air hose having a hose size of 5/16 inch (inner diameter). The air supply line shall be equipped with a filter, air pressure regulator and lubricator.

4.4 First article inspection. When specified, the contractor shall furnish one sample of a completely fabricated tool for first article inspection. The sample shall have been manufactured and inspected using the same materials, processes, and methods that will be used in the production of the tools used to fulfill the terms of the contract or purchase order. The sample shall meet the requirements of this Commercial Item Description. Approval of the first article sample shall not relieve the contractor of the responsibility to furnish tools in accordance with the requirements of this Commercial Item Description. First article inspection shall consist of the following evaluation in the sequence listed: Inspection, Air consumption test, Performance test, Sound pressure level test, Practical service test, Service test, and Examination after service test.

4.5 Production lot inspection and testing: Production lot testing shall consist of the following evaluations conducted in the sequence listed: Inspection, Air consumption test, Speed test, and Practical service test. A sample lot shall consist of all items of same NSN offered for delivery at one time. Sampling procedures shall be in accordance with ANSI/ASQC Z 1.4 at an Inspection Level of S-3. The acceptable quality level (AQL) shall be 1.5 defects per 100 units tested. A sample unit shall consist of one completely fabricated tool.

4.6 Air consumption test: Air consumption shall be measured with either a certified liquid displacement meter or a flowmeter.

4.7 Speed test: Each sample tool shall be tested to determine compliance with the speed requirement specified herein. A stroboscope, tachometer or other suitable device, of sufficient capacity and accuracy, shall be used.

4.8 Sound pressure level test: The sound pressure level test shall be conducted in accordance with the applicable provisions of the Compressed Air and Gas Institute's, "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment". The sound pressure level shall not exceed 90 decibels on the A weighted scale.

4.9 Performance test. A 10 pound weight shall be placed on top of the sander. The sander shall be operated using 80 grit sandpaper on a pine wood surface. The sander shall operate smoothly without any excessive vibration. The sander load speed shall be measured using a stroboscope or equivalent device. The load speed shall not be less than 80 percent of the no-load speed.

4.10 Practical service test: Each sample tool shall be operated for 5 minutes. The test shall duplicate a typical work application. During the test, the tool shall be checked for throttle operation, rotation, excessive noise or vibration, ease of operation, and general serviceability.

4.11 Service test: A simulated service test shall be conducted for 150 hours. A test sample shall be installed in a fixture in a manner that the tool can be loaded and automatically actuated for 150 hours. A load of 5 pounds, in addition to the weight of the tool, shall be applied perpendicular to the plane of the sanding pad during the load portion of the test cycle. Each cycle shall be 2 minutes ON and 2 minutes OFF.

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4.12 Examination after service test: Tools satisfactorily completing the Service test shall be subjected to the Sound pressure level test, Air consumption test, and shall conform to the applicable requirements of these tests. Upon completion of these tests, the tool shall be disassembled and the parts examined. Any tool having broken, cracked, or deformed parts shall be rejected.

5. Preservation, packaging, packing, and marking: Tools shall be preserved, packaged, packed, and marked as specified in the contract or purchase order.

5.1 Identification markings: Each item (including all items within a set or kit) shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

6. Notes:

6.1 The issue of referenced documents in effect on the date of invitation for bids or request for proposal shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the Contracting Officer.

6.2 Military standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.

6.3 ANSI standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.4 The "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment" is available from the Compressed Air and Gas Institute, 1300 Sumner Avenue, Cleveland, OH 44115.

Military Coordinating Activity

Army - GL

Preparing Activity

GSA - FSS

Custodians

Army - GL
Air Force - 99

Review Activities

Air Force - 35, 84

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